Ball-by-Ball Fielding Impact Evaluation

A Data-Driven Analysis of Fielding Performance

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Contents

1 Introduction **2**

1.1 Motivation 2

1.2 Problem Statement 2

2 Data Collection and Structure **3**

2.1 Match Context 3

2.2 Ball-by-Ball Dataset 3

2.2.1 Key Columns 3

3 Data Cleaning and Preparation **4**

3.1 Numeric Normalization 4

3.2 Categorical Standardization 4

4 Fielding Impact Scoring Framework **5**

4.1 Design Philosophy 5

4.2 Scoring Components 5

4.3 Per-Ball Fielding Score 5

5 Aggregation and Final Metrics **6**

5.1 Fielder-Level Aggregation 6

5.2 Normalized Fielding Rating 6

6 Results and Visual Analysis **7**

6.1 Fielding Impact Score by Player 7

6.2 Runs Saved vs Errors 7

6.3 Overall Fielding Distribution 8

7 Conclusion and Future Scope **9**

7.1 Key Contributions 9

7.2 Future Extensions 9

7.3 Implementation Notebook (Google Colab) 9

Chapter 1

Introduction

1.1 Motivation

In modern cricket analytics, batting and bowling performances are extensively measured using well-defined statistical metrics such as averages, strike rates, wickets, and economy rates. Fielding, however, remains one of the least quantified aspects of the game and is often judged subjectively through commentary or visual assessment.

Key match-defining moments such as boundary saves, run-outs, diving stops, and misfields directly influence match outcomes, yet they are rarely captured in traditional scorecards. This project was motivated by the need to convert these qualitative fielding contributions into measurable, objective metrics.

1.2 Problem Statement

The objective of this project is to design a transparent, explainable system that evaluates fielding performance on a ball-by-ball basis using structured match data, without relying on complex machine learning or natural language processing techniques.

Chapter 2

Data Collection and Structure

2.1 Match Context

The analysis focuses on a single T20 international match:

- India Women vs Sri Lanka Women
- Match Date: 30 December 2025
- India fielded second and won the match

2.2 Ball-by-Ball Dataset

A custom CSV dataset was constructed where each row represents one delivery in the innings. The dataset includes both cricketing context and explicit fielding involvement.

2.2.1 Key Columns

- **over_ball**: Ball number (e.g., 8.6)
- **fielder**: Player involved in the fielding action
- **fielding_action**: Catch, run-out, stop, misfield, or none
- **runs_saved**: Runs prevented due to fielding
- **error_runs**: Runs conceded due to fielding errors
- **is_boundary_prevented**: Indicates boundary prevention
- **effort_level**: Qualitative effort rating

Missing information was intentionally allowed and handled later during processing.

Chapter 3

Data Cleaning and Preparation

Raw CSV data often contains missing values, inconsistent data types, and mixed formats. Before analysis, the dataset was cleaned using Python.

3.1 Numeric Normalization

Columns such as runs saved and error runs were converted into numeric format. Invalid or missing entries were safely converted to zero to ensure mathematical stability.

3.2 Categorical Standardization

Categorical columns such as fielding actions and effort levels were standardized to avoid ambiguity during scoring. This ensured consistency across all deliveries.

This step was crucial in preventing calculation errors and ensuring reproducibility.

Chapter 4

Fielding Impact Scoring Framework

4.1 Design Philosophy

The scoring system was designed using intuitive cricket logic:

- Positive actions increase a fielder's score
- Errors reduce the score
- Effort is rewarded but does not overpower outcomes

4.2 Scoring Components

- **Catches**: +10 points
- **Run-outs**: +12 points
- **Runs Saved**: +1 per run
- **Boundary Prevention**: +2 points
- **Missed Catch**: 8 points
- **Misfield Error**: error runs
- **Effort Bonus**: 0 to +2 points

4.3 Per-Ball Fielding Score

For each delivery, the fielding score is calculated as:

$$\text{Fielding Score}_{ball} = \text{Action Points} + \text{Runs Saved} + \text{Boundary Bonus} - \text{Error Runs} - \text{Missed Catch Penalty}$$

Chapter 5

Aggregation and Final Metrics

5.1 Fielder-Level Aggregation

All per-ball scores were aggregated by fielder to compute:

- Total runs saved
- Total error runs
- Number of catches and run-outs
- Total effort bonus
- Final fielding impact score

5.2 Normalized Fielding Rating

To enable intuitive comparison, scores were normalized to a 0–10 scale, where the top-performing fielder in the match receives a rating of 10. I even thought of applying pressure and momentum metrics but the data we have is less so went with this approach instead

Chapter 6

Results and Visual Analysis

6.1 Fielding Impact Score by Player

Figure 6.1 ranks players by their overall fielding contribution.

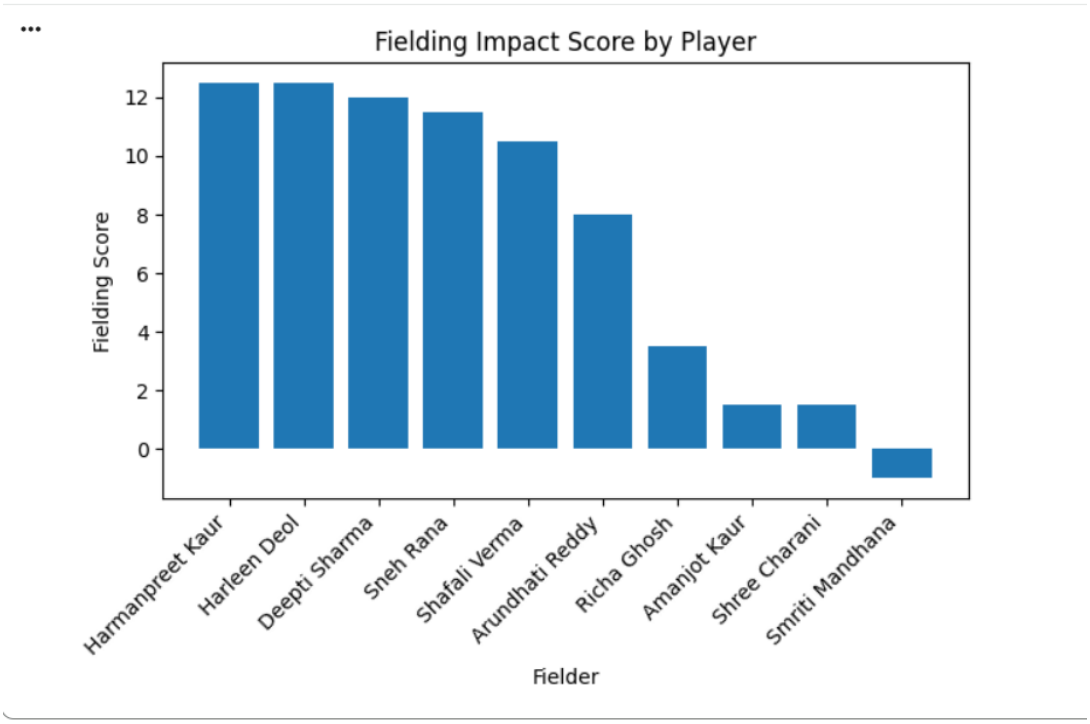

Figure 6.1: Fielding Impact Score by Player

6.2 Runs Saved vs Errors

Figure 6.2 highlights the balance between positive and negative fielding actions.

A1 | fielder

	A	B	C	D	E	F	G	H	I	J
1	fielder	balls_invol	total_runs	total_erro	catches	run_outs	boundary_	total_effo	fielding_sc	fielding_rating_10
2	Harmanpreet Kaur	1	0	0	0	1	0	0.5	12.5	10
3	Harleen Deol	2	1	0	1	0	0	1.5	12.5	10
4	Deepti Sharma	1	0	0	1	0	0	2	12	9.6
5	Sneh Rana	4	4	0	0	0	1	5.5	11.5	9.2
6	Shafali Verma	1	0	0	1	0	0	0.5	10.5	8.4
7	Arundhati Reddy	2	2	0	0	0	1	4	8	6.4
8	Richa Ghosh	3	2	0	0	0	0	1.5	3.5	2.8
9	Amanjot Kaur	2	0	0	0	0	0	1.5	1.5	1.2
10	Shree Charani	1	1	0	0	0	0	0.5	1.5	1.2
11	Smriti Mandhana	1	0	1	0	0	0	0	-1	-0.8
12										

Figure 6.2: Runs Saved vs Errors by Fielder

6.3 Overall Fielding Distribution

Figure 6.3 shows the overall spread of fielding contributions across players.

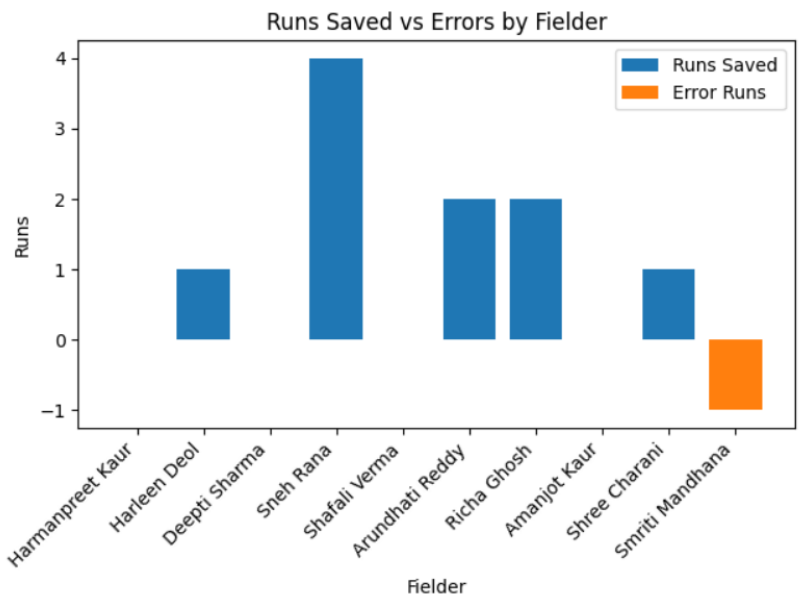

Figure 6.3: Overall Fielding Contribution Distribution

Chapter 7

Conclusion and Future Scope

7.1 Key Contributions

This project demonstrates that fielding performance can be objectively quantified using:

- Ball-by-ball analysis
- Transparent scoring rules
- Error-aware and effort-aware evaluation

The framework bridges the gap between qualitative commentary and quantitative analytics.

7.2 Future Extensions

Possible extensions include:

- Multi-match and tournament-level fielding ratings
- Position-based difficulty weighting
- Integration with video tracking data
- Team-level fielding efficiency metrics

7.3 Implementation Notebook (Google Colab)

All data processing, metric computation, and visualization for this project were implemented using Python in a Google Colab environment.

The complete, executable notebook — including:

- Data loading and cleaning

- Ball-by-ball fielding event processing
- Fielding impact score calculation
- Player-level aggregation
- Visualization generation

is publicly available at the following link:

`https://colab.research.google.com/drive/1g4JrerOVO6x35GTCgGrxD3Z3RxJ8aI7d`

This notebook serves as the reproducible implementation of the methodology described in this report.